

Class #5 - Complex eigenvalues



1. Example: Let the matrix $A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$ acts on $\mathbb{C}^2$. Find the eigenvalues and a basis for each eigenspace in $\mathbb{C}^2$.

$\text{Nul}(A - \lambda I)$ λ : eigenvalue of A
eigenspace in $\mathbb{C}^2$

$$A \vec{x} = \lambda \vec{x}$$

real
complex

$$\vec{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\vec{x} \in \mathbb{C}^2$

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & -2 \\ 1 & 3-\lambda \end{vmatrix}$$

$$= (1-\lambda)(3-\lambda) + 2$$

$$= 5 - 4\lambda + \lambda^2 = 0$$

$$\lambda = \frac{4 \pm \sqrt{16 - 4 \cdot 1 \cdot 5}}{2} = \frac{4 \pm \sqrt{-4}}{2} = 2 \pm i$$

Complex number: $a + ib$

$a \in \mathbb{R}, b \in \mathbb{R}$

Any real number is also a complex number by taking $b=0$.

Real number $\subseteq$ Complex number

Because $\lambda_1 = 2 + i$ and $\lambda_2 = 2 - i$ are two distinct eigenvalues, of each alg. mult. = 1, geometric mult. of each eigenvalue = 1.

$$\text{Nul}(A - (2+i)I)$$

$$\begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} 2+i & 0 \\ 0 & 2+i \end{bmatrix}$$

$$= \begin{bmatrix} -1-i & -2 \\ 1 & 1-i \end{bmatrix}$$

$$r_2 \leftarrow r_2 + \frac{1}{-1-i} r_1$$

$$\sim \begin{bmatrix} 1 & 1-i \\ -1-i & -2 \end{bmatrix}$$

$$r_2 \leftarrow r_2 + (1+i)r_1$$

Eigenspace corresponding to $\lambda = 2+i$ is

$$\text{Span} \left\{ \begin{bmatrix} 1-i \\ -1 \end{bmatrix} \right\}$$

$$\text{Basis is } \left\{ \begin{bmatrix} 1-i \\ -1 \end{bmatrix} \right\}$$

$$\text{Nul}(A - (2-i)I)$$

2nd col $\subseteq \text{span}\{\text{1st col}\}$

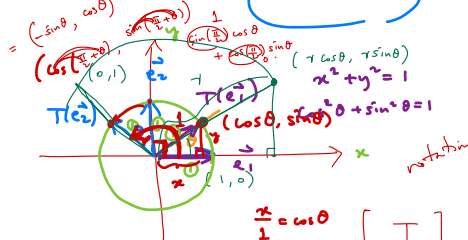
$$= (1-i)(\text{1st col})$$

$$= -2 + (1-i)(-1)$$

$$= -2 + 1 - (-1)$$

$$= 0$$

2. The transformation $x \mapsto Ax$ is the composition of a rotation and a scaling. Give the angle θ of the rotation, where $-\pi < \theta \leq \pi$, and give the scale factor γ .



$$A = \begin{bmatrix} \sqrt{3} & 3 \\ -3 & \sqrt{3} \end{bmatrix}$$

Standard Matrix for T

$$= \begin{bmatrix} T(\vec{e}_1) & T(\vec{e}_2) \end{bmatrix}$$

$$= \begin{bmatrix} T \begin{bmatrix} 1 \\ 0 \end{bmatrix} & T \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Standard Matrix

Rotation mapping

$$\begin{bmatrix} \gamma \cos \theta & -\gamma \sin \theta \\ \gamma \sin \theta & \gamma \cos \theta \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{3} & 3 \\ -3 & \sqrt{3} \end{bmatrix}$$

Find γ and θ

$$\theta = -\pi/3$$

$$\gamma = \sqrt{12} = 2\sqrt{3}$$

3+53: Note: If $A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, where a and b are real and not both zero, then the eigenvalues are $\lambda = a \pm ib$. If we let $\gamma = |\lambda|$, then $\sqrt{a^2 + b^2} = |\lambda|$.

The angle θ is called the *argument* of $\lambda = a + bi$.

$$A = \gamma \begin{bmatrix} a/\gamma & -b/\gamma \\ b/\gamma & a/\gamma \end{bmatrix}$$

$$= \begin{bmatrix} |\lambda| & 0 \\ 0 & |\lambda| \end{bmatrix} \begin{bmatrix} a/|\lambda| & -b/|\lambda| \\ b/|\lambda| & a/|\lambda| \end{bmatrix}$$

(Diagonal Matrix) (Rotation) Factorization
Where

$$\lambda_1 = a + ib$$

$$\lambda_2 = a - ib$$

4. It is true that for any real 2×2 matrix with a complex eigenvalue, it is actually a composition of rotation and scaling after an appropriate change of variables.

5. Theorem: Let A be a real 2×2 matrix with a complex eigenvalue $\lambda = a - bi$ ($b \neq 0$) and an associated eigenvector v in $\mathbb{C}^2$. Then,

$A = PCP^{-1}$, where $P = \begin{bmatrix} \text{Re } v & \text{Im } v \end{bmatrix}$ and $C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$.

A is similar to C .

Proof:

(1) For any vector x in $\mathbb{C}^2$, $\text{Re}(Ax) = A(\text{Re } x)$, and $\text{Im}(Ax) = A(\text{Im } x)$.

$$\vec{x} = \begin{bmatrix} x_1 + iy_1 \\ x_2 + iy_2 \end{bmatrix} \quad A\vec{x} = \begin{bmatrix} a & \beta \\ \gamma & \delta \end{bmatrix} \begin{bmatrix} x_1 + iy_1 \\ x_2 + iy_2 \end{bmatrix} = \begin{bmatrix} ax_1 + idy_1 + \beta x_2 + i\beta y_2 \\ \gamma x_1 + i\gamma y_1 + \delta x_2 + i\delta y_2 \end{bmatrix} = \begin{bmatrix} ax_1 + \beta x_2 \\ \gamma x_1 + \delta x_2 \end{bmatrix} + i \begin{bmatrix} dy_1 + \beta y_2 \\ \gamma y_1 + \delta y_2 \end{bmatrix}$$

Let $\vec{v} = \begin{bmatrix} \text{Re } \vec{v} & \text{Im } \vec{v} \end{bmatrix} \in \mathbb{R}^{2 \times 2}$

Then, P is invertible.

Suppose not.

Suppose $\text{Im } \vec{v} = c \text{Re } \vec{v}$.

$$\begin{aligned} \Rightarrow \vec{v} &= \text{Re } \vec{v} + i \text{Im } \vec{v} \\ &= \text{Re } \vec{v} + i(c \text{Re } \vec{v}) \\ &= (1 + ic) \text{Re } \vec{v} \end{aligned}$$

$$A\vec{v} = \lambda \vec{v}$$

$$A(1 + ic) \text{Re } \vec{v} = \lambda (1 + ic) \text{Re } \vec{v}$$

$$\lambda = a - bi$$

$$\lambda = a + bi$$

$$\therefore \text{Re } \vec{v} = \vec{0}$$

$\vec{v}$ is an eigenvector corresponding to $\lambda = a - ib$.
 $\vec{v} \neq \vec{0}$.

$$\therefore \text{Im } \vec{v} = \vec{0}$$

$$\therefore \vec{v} = \vec{0}$$

This is a contradiction.

① Q#3

② 5 min. break

9/25

③ Attendance

④ Participation Quiz

⑤ Class #5 #3 ~

⑥ Will discuss Q#2 Aug. 8-67

⑦ Q#4 Mon. 10/12 PP # 8 ~ 10

① Symmetric matrix $A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ $A^T = A$

② $A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ $a, b \in \mathbb{R}$

eigenvalues of A

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} a-\lambda & -b \\ b & a-\lambda \end{vmatrix} = 0$$

$$\Leftrightarrow (a-\lambda)^2 + b^2 = 0 \quad a^2 - 2a\lambda + \lambda^2 + b^2 = 0$$

$$\Leftrightarrow \lambda = \frac{2a \pm \sqrt{4a^2 - 4(a^2 + b^2)}}{2} = a \pm bi$$

$$A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} A^T$$

"skew-symmetric"

③ $A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} = \underbrace{\begin{bmatrix} \gamma & 0 \\ 0 & \gamma \end{bmatrix}}_{\text{scalar}} \begin{bmatrix} \frac{a}{\gamma} & -\frac{b}{\gamma} \\ \frac{b}{\gamma} & \frac{a}{\gamma} \end{bmatrix}$ eigenvalues of A

$$\gamma = \sqrt{a^2 + b^2}$$

$$\left(\frac{a}{\gamma}\right)^2 + \left(\frac{b}{\gamma}\right)^2 = 1$$

$$\left|\frac{a}{\gamma}\right| \leq 1, \left|\frac{b}{\gamma}\right| \leq 1$$

$$\begin{bmatrix} \frac{a}{\gamma} & -\frac{b}{\gamma} \\ \frac{b}{\gamma} & \frac{a}{\gamma} \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

"rotation"

(2) $A(\operatorname{Re} v) = a\operatorname{Re} v + b\operatorname{Im} v$ and $A(\operatorname{Im} v) = -b\operatorname{Re} v + a\operatorname{Im} v$

$$\begin{aligned}
 A \vec{v} &= \lambda \vec{v} \\
 b \neq 0 &= (a-bi) \left(\begin{bmatrix} \operatorname{Re} \vec{v} \\ \operatorname{Im} \vec{v} \end{bmatrix} + i \begin{bmatrix} \operatorname{Im} \vec{v} \\ -\operatorname{Re} \vec{v} \end{bmatrix} \right) \\
 A \left(\begin{bmatrix} \operatorname{Re} \vec{v} \\ \operatorname{Im} \vec{v} \end{bmatrix} + i \begin{bmatrix} \operatorname{Im} \vec{v} \\ -\operatorname{Re} \vec{v} \end{bmatrix} \right) &= (a-bi) \left(\begin{bmatrix} \operatorname{Re} \vec{v} \\ \operatorname{Im} \vec{v} \end{bmatrix} + i \begin{bmatrix} \operatorname{Im} \vec{v} \\ -\operatorname{Re} \vec{v} \end{bmatrix} \right) \\
 &= a(\operatorname{Re} \vec{v}) - ib(\operatorname{Re} \vec{v}) + ia(\operatorname{Im} \vec{v}) + b(\operatorname{Im} \vec{v}) \\
 A(\operatorname{Re} \vec{v}) + iA(\operatorname{Im} \vec{v}) &= a(\operatorname{Re} \vec{v}) - ib(\operatorname{Re} \vec{v}) + ia(\operatorname{Im} \vec{v}) + b(\operatorname{Im} \vec{v}) \\
 \operatorname{Re}(A\vec{x}) &= A(\operatorname{Re} \vec{x}) \\
 \operatorname{Im}(A\vec{x}) &= A(\operatorname{Im} \vec{x}) \\
 A\vec{x} &= \operatorname{Re}(A\vec{x}) + i\operatorname{Im}(A\vec{x})
 \end{aligned}$$

(3) Conclude $AP = PC$.

$$\begin{aligned}
 A \begin{bmatrix} \operatorname{Re} \vec{v} & \operatorname{Im} \vec{v} \end{bmatrix} &\stackrel{?}{=} \begin{bmatrix} \operatorname{Re} \vec{v} & \operatorname{Im} \vec{v} \end{bmatrix} \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \\
 &= \begin{bmatrix} A \begin{bmatrix} \operatorname{Re} \vec{v} \\ \operatorname{Im} \vec{v} \end{bmatrix} \end{bmatrix} \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \\
 &= \begin{bmatrix} A \begin{bmatrix} \operatorname{Re} \vec{v} \\ \operatorname{Im} \vec{v} \end{bmatrix} \end{bmatrix} \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \\
 &= \begin{bmatrix} A \begin{bmatrix} \operatorname{Re} \vec{v} \\ \operatorname{Im} \vec{v} \end{bmatrix} \end{bmatrix} \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \\
 &= \begin{bmatrix} a \operatorname{Re} \vec{v} + b \operatorname{Im} \vec{v} & -b \operatorname{Re} \vec{v} + a \operatorname{Im} \vec{v} \end{bmatrix} \\
 &\quad \text{True! by (2)}
 \end{aligned}$$

6. Example: $A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$